Wayne Williams Opanja

Software & AI Engineer · Builds production AI products end to end

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SELECTED IMPACT

- **Shipped national govtech:** led frontend and UX for Nigeria's Federal Ministry of Education RICC platform, now live at ricc.education.gov.ng, and worked on the federal Trademark Registry.
- **Put AI to work for a county government:** built a WhatsApp assistant that lets Nairobi City County revenue staff ask about collections in plain words and get answers from the county's own records. The AI can read the data but never change it.
- **Turned a climate-risk framework into software banks can use:** built British International Investment's climate and nature risk toolkit, so financial institutions can screen clients, measure financed emissions and stress-test portfolios step by step.
- **Delivered for clients in 6 countries:** 8 projects for 8 clients at Pneuma Labs across the UK, Kenya, Uganda, Rwanda, Burundi and the DRC.

SUMMARY

Software and AI engineer who takes products from the first discovery call to production. I build LLM agents, RAG and text-to-SQL systems, WhatsApp-first products and full-stack web apps in TypeScript and Python, and I put guardrails around AI so it can be trusted with real data. Background in data science, climate-risk analytics and security (CHFI). Fluent in English and Swahili.

CORE SKILLS

AI: LLM agents and multi-agent pipelines, RAG, text-to-SQL, tool calling, provider fallback, output validation, OCR and document parsing, scikit-learn, PyTorch.

Engineering: TypeScript, JavaScript, Python, Java, PHP; Next.js, React, Node.js, NestJS, Express, Django, FastAPI; PostgreSQL, Supabase, Prisma, MongoDB; REST APIs, Kafka.

Delivery: Docker, Kubernetes, GitHub Actions, Vercel, Render, AWS, Azure, GCP; Vitest, Jest, pytest, Playwright; WhatsApp, Twilio, Paystack, Smile ID, Sumsub.

Product: discovery and user research, requirements, UX and prototyping, QA and UAT, client and stakeholder management.

EXPERIENCE

Lead Software Engineer · Pneuma Labs | Nairobi, clients in 6 countries | 2026 – Present

AI and automation company: 8 projects for 8 clients in logistics, real estate, mining, financial services and government.

- **Nairobi City County revenue assistant:** turned the county's revenue reports into one clean database and added an AI assistant staff can question on WhatsApp or the web. Checked its answers against the original reports and fixed double-counting before staff relied on it. *Next.js, Supabase/PostgreSQL, LLMs, WhatsApp.*
- **NSE market research assistant:** built a team of AI agents that research Nairobi Securities Exchange stocks every morning, argue the case for and against, check risk against inflation and log each call so its accuracy can be measured. Used by analysts in Kenya. *TypeScript, multi-agent LLMs, Supabase, GitHub Actions.*
- **PropFlow property-management platform:** designed and built one system for landlords to run rent, leases, repairs and payments, with a WhatsApp line for tenants and emergency phone calls. AI suggests the next step; rules decide before anything happens. *Next.js, Prisma, PostgreSQL, Twilio, Paystack, Docker.*
- **Process automation:** mapped operations for Mediscope and the Strathmore Centre for Value Investing and designed automations around how their teams work.

Consultant, Climate & Nature Toolkit · British International Investment | Remote | 2026 – Present

- **Made a development-finance framework usable:** built the platform behind BII's climate and nature risk toolkit for financial institutions: client climate screening, maturity assessment, financed-emissions and stress-test calculators, with PDF, Excel and Word exports (Next.js, TypeScript, Prisma, PostgreSQL).

Senior Software Engineer · ProDG Studios | Remote, UK-based | 2025 – 2026

- **RICC, Federal Ministry of Education, Nigeria:** led frontend and UI/UX. Researched why universities, industry and regulators weren't connecting and turned that into the user journeys for innovation assessment and routes to funding and IP support.
- **Trademark Registry, Federal Ministry of Industry, Trade and Investment, Nigeria:** built digital registration and administration workflows that replaced manual processes.
- **CrossVest, Venture Garden Group:** developed and deployed the frontend of an investment marketplace that partner apps can embed so people in the diaspora can invest in vetted African opportunities. Took it through user acceptance testing into production, with identity checks through Smile ID and Sumsub.
- **NGAO, Action Lab:** built a WhatsApp-first AI data-collection platform that turns conversational answers into structured records ready for analysis.

Data Scientist, then Partner · Climate Shapers Africa | Nairobi | 2025 – Present

- **Climate risk for banks:** built climate-stress exposure analytics for the Kenya and Tanzania Bankers Associations, linking climate indicators to probability of default, with NCBA, Equity Bank and Victoria Commercial Bank among the stakeholders.
- **Data products:** built the Africa Climate & Nature Data Platform prototype (sector dashboards and a Kenya county risk map), Power BI dashboards and AI research automations for the Kenya Climate & Nature Directory. Partner since 2026.

Earlier experience

- **Strathmore Institute of Mathematical Sciences, Student Assistant (Developer) (2023 – 2025):** built a course evaluation system (Django, React, PostgreSQL) that raised student feedback rates by 75%.
- **Harambee SACCO, Penetration Testing Consultant (2024):** tested a financial cooperative's systems and wrote up findings for remediation.
- **Strathmore University ICT, Software Developer Intern (2025 – 2026, degree internship):** rebuilt the phone-billing system as an Odoo module with call-record and HR/ERP sync.

SELECTED PROJECTS

- **MessageHeard v3 (team project, 2025 – 2026):** worked on a multi-tenant AI video-generation platform with Clerk auth, role-based access, transcript-to-scene breakdown and a job queue for concurrent image and video generation.
- **Chat Your Finances (lead developer):** AI assistant that reads M-Pesa, bank and payslip statements and answers money questions on WhatsApp, Telegram and web, with a KRA tax-filing guide. *Next.js, Supabase, RAG, OCR.*

EDUCATION AND CERTIFICATIONS

- **Strathmore University:** Bachelor of Business Information Technology (2024 – 2027, in progress, Grade A average); Diploma in Business Information Technology (2023 – 2024).
- **Certifications:** CHFI, EC-Council (2024); Applied Data Science & ML, WorldQuant University (2025); Google Cybersecurity; Simplilearn Full-Stack Development.
- **Languages:** English and Swahili (fluent), German (beginner).